Mathematical Background for `AccumulatR`

Niek Stevenson

Abstract

This note gives a standalone mathematical description of the framework implemented in `AccumulatR`. The package is built from a small number of objects: leaf-level completion times, threshold pools, logical outcome rules, timing modifiers such as chained onsets and triggers, and an observation layer for mixtures, remapped labels, and ranked responses. The aim here is to keep the notation minimal and to build the full model in layers, starting from a simple race and then adding the special cases that the package supports.

1 Overview

The framework can be read in four layers.

1. A *leaf* is a single latent finishing-time process.
2. A *source* is either a leaf or a threshold pool built from other sources.
3. An *outcome rule* says when a response should count as having happened. Rules are built from sources with helpers such as `first_of()`, `all_of()`, `none_of()`, and `inhibit()`.
4. An *observation model* turns latent outcome rules into the data we actually record: one winner, several ranked winners, a mixture over latent components, or relabelled observed responses.

The important point is that the package does not need a different likelihood formula for every named model. It always starts from leaf completion times and then composes them.

2 Leaf Times

Let a index leaves. Each leaf has a completion time T_a and corresponding distribution functions

$$F_a(t) = \Pr(T_a \le t), \qquad f_a(t) = \frac{d}{dt} F_a(t), \qquad S_a(t) = 1 - F_a(t).$$

We use the same notation for any other object in the model: once a pool or an outcome rule has a completion time, it also has F, f, and S.

At the leaf level, `AccumulatR` only needs a valid density–CDF pair. The current implementation supplies the families

`lognormal`, `gamma`, `exgauss`, `LBA`, `RDM`.

Everything else in the framework is distribution-agnostic.

Absolute onset. If a leaf has an absolute onset $d_a \ge 0$ and a baseline finishing time $T_a^\star$, then

$$T_a = d_a + T_a^\star.$$

Equivalently,

$$F_a(t) = F_a^\star(t - d_a), \qquad f_a(t) = f_a^\star(t - d_a),$$

with both terms equal to 0 for $t \le d_a$.

3 Pools

A pool P has a member set $M(P)$ and a threshold k_P. It finishes when the k_Pth member finishes. Hence

$$T_P = \text{the } k_P\text{th order statistic of } \{T_m : m \in M(P)\}.$$

The simplest description of the pool CDF is

$$F_P(t) = \Pr\left(\sum_{m \in M(P)} \mathbf{1}\{T_m \le t\} \ge k_P \right). \tag{1}$$

For heterogeneous independent members, the density is

$$f_P(t) = \sum_{j \in M(P)} f_j(t) \sum_{\substack{A \subseteq M(P) \setminus \{j\} \\ |A| = k_P - 1}} \prod_{a \in A} F_a(t) \prod_{b \in M(P) \setminus (A \cup \{j\})} S_b(t). \tag{2}$$

Three special cases matter often:

- $k_P = 1$: the pool is a *first finisher*.
- $k_P = |M(P)|$: the pool is a *last finisher*.
- $|M(P)| = 1$: the pool is just an alias for one source.

Pools can be nested. Once the member channels of a source are known, the same definitions apply whether the members are leaves or other pools.

4 Outcome Rules

An outcome rule is an expression built from sources. For any rule E, let T_E be the time at which the rule first becomes true, with distribution functions F_E, f_E, and S_E.

4.1 Direct Event Rules

If the rule is just a source x, then

$$T_E = T_x, \qquad F_E = F_x, \qquad f_E = f_x.$$

This is the usual simple race-model case.

4.2 `first_of()`

If

$$E = \texttt{first_of}(E_1, \ldots, E_m),$$

then the rule becomes true when the first child becomes true, so

$$T_E = \min(T_{E_1}, \ldots, T_{E_m}).$$

Hence

$$F_E(t) = 1 - \prod_{r=1}^{m} S_{E_r}(t), \qquad f_E(t) = \sum_{r=1}^{m} f_{E_r}(t) \prod_{q \neq r} S_{E_q}(t). \tag{3}$$

4.3 `all_of()`

If

$$E = \texttt{all_of}(E_1, \ldots, E_m),$$

then the rule becomes true when the last required child becomes true, so

$$T_E = \max(T_{E_1}, \ldots, T_{E_m}).$$

Hence

$$F_E(t) = \prod_{r=1}^{m} F_{E_r}(t), \qquad f_E(t) = \sum_{r=1}^{m} f_{E_r}(t) \prod_{q \neq r} F_{E_q}(t). \tag{4}$$

4.4 `none_of()`

`none_of()` is best read as an *absence condition*. If a rule requires `none_of(B)`, then by time t the relevant quantity is

$$\Pr(B \text{ has not happened by } t) = S_B(t).$$

For example, if a response is defined as

$$\texttt{all_of}(A, \texttt{none_of}(B)),$$

then the rule holds by time t with probability $F_A(t)S_B(t)$.

So `none_of()` does not create a new race winner. It contributes a survival factor.

4.5 `inhibit()`

The guard rule is the package's way to write a stop-like blocker. Let

$$G = \texttt{inhibit}(A, B; U_1, \ldots, U_m),$$

where A is the reference event, B is the blocker, and $U_1, \ldots, U_m$ are optional `unless` events that can neutralize the blocker.

Define the guard-survival factor

$$H_G(t) = 1 - \int_0^t f_B(u) \prod_{r=1}^{m} S_{U_r}(u)\, du. \tag{5}$$

This is the probability that by time t the blocker has *not* occurred at a time when all rescue events were still absent.

The guarded rule then has density

$$f_G(t) = f_A(t)\, H_G(t), \qquad F_G(t) = \int_0^t f_G(u)\, du. \tag{6}$$

The common special case has no `unless` terms. Then

$$H_G(t) = S_B(t), \qquad f_G(t) = f_A(t)S_B(t).$$

This is the standard "go finishes at t and stop has not finished yet" construction.

5 From Outcome Rules to Trial Likelihood

Let the model define outcome rules $R_1, \ldots, R_J$. For an observed response r at time t, the fundamental quantity is

$$\ell_r(t) = \frac{d}{dt} \Pr\big(T_{R_r} \le t,\ T_{R_j} > t \text{ for all } j \ne r\big)\,. \tag{7}$$

5.1 Simple disjoint races

If the outcome rules are disjoint, Equation (7) reduces to the familiar race-model formula

$$\ell_r(t) = f_{R_r}(t) \prod_{j \ne r} S_{R_j}(t). \tag{8}$$

This is exactly the likelihood for the usual two-choice race, for a pooled response against a competitor, or for any other case where the outcome rules do not overlap.

5.2 Overlapping outcome rules

`AccumulatR` also allows outcomes whose definitions overlap. In that case the correct object is not a hand-made tie-breaking rule. It is an inclusion–exclusion calculation over overlapping outcome events.

Let $\mathcal{C}_r = \{1, \ldots, J\} \setminus \{r\}$ be the competitor set for outcome r. Then

$$\ell_r(t) = \sum_{A \subseteq \mathcal{C}_r} (-1)^{|A|} f_{r \wedge A}(t), \tag{9}$$

where $f_{r \wedge A}(t)$ is the density of the conjunction

$$R_r \wedge \bigwedge_{j \in A} R_j$$

becoming true at time t.

This is the mathematical core of overlap handling in the exact back-end. Under continuous-time leaf distributions, exact equal-time ties have probability zero; the real issue is overlapping *definitions*, and that is what Equation (9) resolves.

No-response mass. Some models leave positive probability on "no observed response", for example because a trigger fails. Before any relabelling or guessing, that probability is

$$p_{\text{none}} = 1 - \sum_{r=1}^{J} \int_0^\infty \ell_r(t)\, dt. \tag{10}$$

6 Timing Modifiers

6.1 Chained onsets

If a leaf a starts only after another source u has finished, with an additional lag $\delta \geq 0$, then

$$T_a = T_u + \delta + T_a^\star,$$

where $T_a^\star$ is the leaf's own post-start finishing time.

The resulting density is a convolution:

$$f_a(t) = \int_0^{t-\delta} f_u(s)\, f_a^\star(t - \delta - s)\, ds. \tag{11}$$

Likewise,

$$F_a(t) = \int_0^\infty F_a^\star(t - \delta - s)\, f_u(s)\, ds. \tag{12}$$

If the onset source u is itself a pool, then f_u and F_u are the pool channels from Equations (1)–(2).

6.2 Triggers

Triggers are start indicators. They are easiest to write with Bernoulli variables.

Independent trigger. If leaf a has trigger success probability $1 - q_a$, let

$$Z_a \sim \text{Bernoulli}(1 - q_a),$$

and set $T_a = \infty$ whenever $Z_a = 0$. Then

$$F_a(t) = (1 - q_a) F_a^\dagger(t), \qquad f_a(t) = (1 - q_a) f_a^\dagger(t), \tag{13}$$

where $(F_a^\dagger, f_a^\dagger)$ are the channels the leaf would have had without trigger failure.

Shared trigger. If a group of leaves shares one trigger, then one Bernoulli variable Z_g controls all of them. Conditional on $Z_g = 1$, all members are available. Conditional on $Z_g = 0$, none of them start. The trial likelihood is then an average over trigger states:

$$L = \sum_z \Pr(Z = z)\, L(z). \tag{14}$$

This is the only place where the package deliberately introduces joint start dependence across several leaves.

7 Observation Layer

7.1 Mixture components

A component c defines a variant model with active leaf set A_c and weight π_c. If the component is observed on a trial, the contribution is simply that component's likelihood:

$$L = L_c.$$

If the component is latent, the trial contribution is

$$L = \sum_c \pi_c L_c, \qquad \sum_c \pi_c = 1. \tag{15}$$

In sampled mixtures one component is taken as the reference component, and its weight is whatever mass remains after the other weights are assigned.

7.2 Outcome remapping and guessing

The semantic outcome generated by the race need not be the same as the observed label. Let M_{yr} be the probability that semantic outcome r is observed as label y. Then

$$\ell_y(t) = \sum_r M_{yr}\, \ell_r(t). \tag{16}$$

This covers simple many-to-one relabelling and probabilistic guessing rules. If the observed label is kept but the response time is discarded, the contribution becomes the integrated mass

$$p_y = \int_0^\infty \ell_y(t)\, dt. \tag{17}$$

7.3 Ranked observations

If the model records the first K outcomes on a trial, the data are

$$(r_1, t_1), \ldots, (r_K, t_K), \qquad 0 < t_1 < \cdots < t_K.$$

In the current implementation, ranked observation is intentionally narrower than the full expression language. It is supported for direct event outcomes with the exact backend, and the declared outcomes must have disjoint deterministic supports.

Under those conditions, the ranked likelihood is the joint density of the ordered completion event

$$L_{\text{rank}} = \frac{\Pr(T_{r_1} \in dt_1, \ldots, T_{r_K} \in dt_K,\ T_j > t_K \text{ for all } j \notin \{r_1, \ldots, r_K\})}{\prod_{m=1}^{K} dt_m}. \tag{18}$$

For independent direct outcomes this simplifies to

$$L_{\text{rank}} = \left[\prod_{m=1}^{K} f_{r_m}(t_m)\right] \left[\prod_{j \notin \{r_1, \ldots, r_K\}} S_j(t_K)\right]. \tag{19}$$

The exact evaluator computes this sequentially by removing already used outcomes after each observed rank.

8 Special Cases

The general framework above contains the common models used in the package.

Simple two-choice race. If R_1 and R_2 are direct outcomes from disjoint leaves,

$$\ell_{R_1}(t) = f_{R_1}(t) S_{R_2}(t), \qquad \ell_{R_2}(t) = f_{R_2}(t) S_{R_1}(t).$$

2-of-3 pooled response. If response A is a 2-of-3 pool and response B is a single competing leaf, first compute (F_A, f_A) from Equations (1) and (2), then use

$$\ell_A(t) = f_A(t) S_B(t), \qquad \ell_B(t) = f_B(t) S_A(t),$$

provided the two outcomes are disjoint.

Stop-like inhibition. If response `go` is written as `inhibit(go, stop)`, then

$$\ell_{\text{go}}(t) = f_{\text{go}}(t) S_{\text{stop}}(t) \prod_{j \neq \text{go}} S_{R_j}(t)$$

in the simple disjoint case.

Staged processing. If leaf C starts only after B has finished, then the response-time distribution for C is the convolution in Equation (11).

Latent fast/slow mixture. If trials arise from a fast component and a slow component with weight π, then

$$L = \pi L_{\text{fast}} + (1 - \pi) L_{\text{slow}}.$$

9 What the Package Currently Supports

The mathematics above matches the current code path in `src/`. The implemented framework includes

- leaf distributions from the five supported families,
- threshold pools, including nested pools,
- outcome rules built from the logical helpers `first_of`, `all_of`, `none_of`, and `inhibit`,
- absolute onsets and chained onsets via `after()`,
- independent and shared triggers,
- fixed and sampled mixtures,
- observed-label remapping and guessing,
- top-K ranked readout subject to the restrictions stated above.

It does *not* need separate mathematical theories for "ranked models", "mixture models", "guard models", and so on. Those are all special cases of the same layered construction:

$$\begin{aligned} \text{leaf times} &\longrightarrow \text{source times} \longrightarrow \text{outcome rules} \\ &\longrightarrow \text{trial likelihood} \longrightarrow \text{observation model.} \end{aligned}$$